

Electrochemical equilibrium

After midterm · Session 6 of 6 · 180 minutes

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Session outcome

Connect balanced half-reactions, activities, potential and Gibbs energy.

Before class: Review reaction stoichiometry, $\Delta_r G$ and dimensionless reaction quotients.

Learning path and all labs

The 180-minute class

IN CLASS	MINUTES
Recall and prediction	10
Concepts and derivation	45
Worked example	30
Break	10
Instructor lab demonstration	25
Guided student exploration	35
Discussion and interpretation	15
Exit question and independent task	10

Chemical and electrochemical potential

For an ion of charge number z_i ,

$$\tilde{\mu}_i = \mu_i + z_i F \varphi$$

Chemical and electrical contributions enter the driving force. The cell potential connects that driving force to electrical work under reversible conditions.

Two reduction half-reactions

Enter both half-reactions as reductions. Reverse the left reaction and add the right after matching electron counts.

$$E_{cell}^{\circ} = E_{right}^{\circ} - E_{left}^{\circ}$$

Both potentials need the same reference electrode and temperature. Multiplying a half-reaction does not multiply its potential.

Electron and charge bookkeeping

A reduction consuming n electrons has

$$\sum_{\text{species}} \nu_i z_i + n = 0$$

Lab 13 excludes electrons from the species table and adds their charge using the entered n . It checks the declared atom counts independently.

Matching electron counts uses their least common multiple.

Potential, Gibbs energy and K

$$\Delta_r G^\circ = -nFE^\circ, \quad \ln K = \frac{nFE^\circ}{RT}$$

$$\Delta_r G = -nFE$$

Positive E favors the written cell reaction. At complete-cell equilibrium $Q=K$ and $E=0$.

These relationships refer to the written reaction basis.

Nernst equation from reaction thermodynamics

Combine $\Delta_r G = \Delta_r G^\circ + RT \ln Q$ with $\Delta_r G = -nFE$:

$$E = E^\circ - \frac{RT}{nF} \ln Q$$

At 298.15 K,

$$E = E^\circ - \frac{0.05915935 \text{ V}}{n} \log_{10} Q$$

Changing T requires valid E° data at the new T.

Activities remain separate by compartment

For identical M^{2+}/M electrodes,



$$Q = \frac{a(M_L^{2+})}{a(M_R^{2+})}, \quad E^\circ = 0$$

The same ion in different compartments cannot be canceled when its activities differ. Pure present solids have unit activity.

A concentration-cell calculation

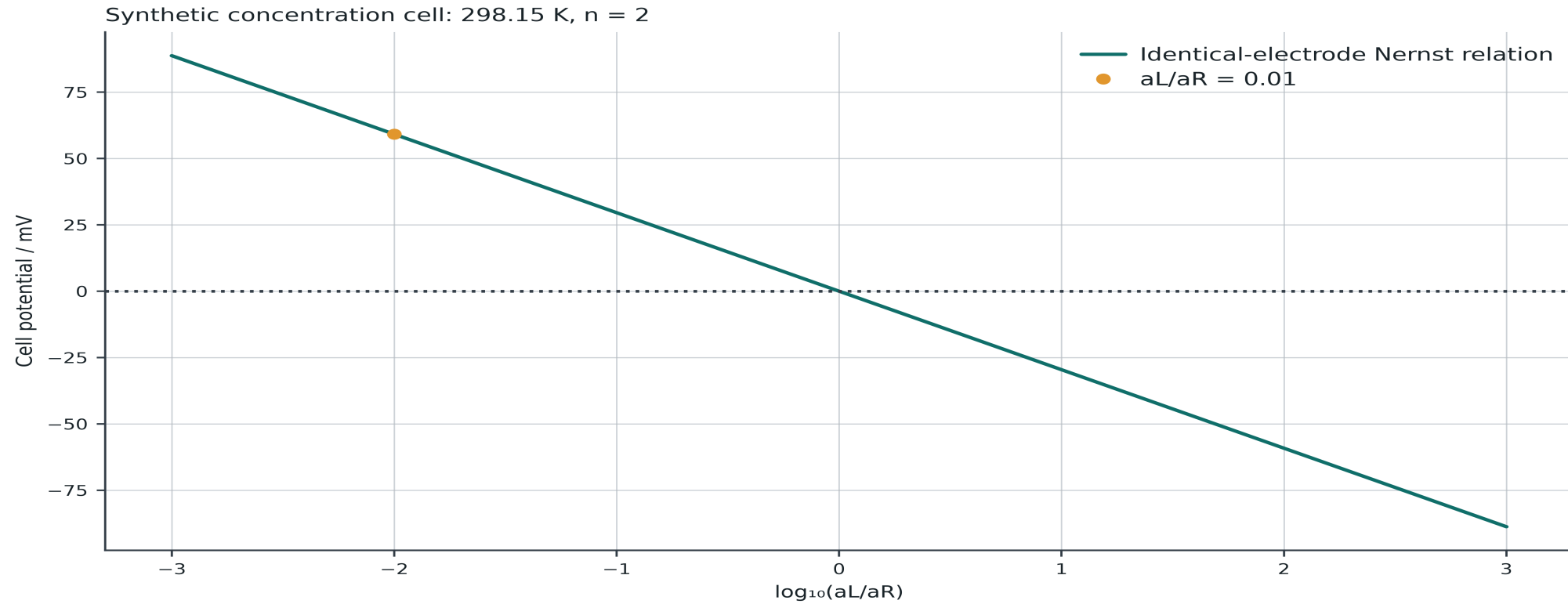
At 298.15 K, $n=2$, $a_L=0.01$ and $a_R=1$:

$$Q = 0.01, \quad E = +0.05915935 \text{ V}$$

$E^\circ=0$, $K=1$, but E is nonzero. Reversing the concentrations changes the sign of E . Equal activities give $E=0$.

$\Delta_r G \approx -11.416 \text{ kJ}$ per mol of the written reaction.

Nernst sensitivity to an activity ratio



Synthetic identical-electrode cell at 298.15 K with $n=2$ and zero junction potential.

Concentration is not activity

For a molarity standard,

$$a_i = \gamma_i c_i / c^\circ$$

Changing γ changes the activity ratio even if both concentrations are unchanged. A molality-based γ cannot be inserted into this equation without a consistent conversion.

Individual-ion activity conventions and reference states must match the potential data.

Reaction scaling leaves E unchanged

Multiply the full cell reaction by two:

QUANTITY	TRANSFORMATION
$n, \Delta rG^\circ, \Delta rG$	Multiply by 2
$\ln K, \ln Q$	Multiply by 2
K, Q	Square
E°, E	Unchanged

Potential is a driving force per unit charge, not a reaction energy.

Class demonstration • Lab 13

Open electrochemistry

1. Load the concentration-cell preset and predict E.
2. Swap activities; then make them equal.
3. Change an electron count incorrectly and inspect the balance rejection.
4. Restore the valid study export and reproduce the result.

Guided exploration • 35 minutes

Produce two concentration-cell states and a reaction-scaled version of one state.

Check $E_{\text{right}} - E_{\text{left}}$, $-nFE$ and the stoichiometric/charge balances. Explain any changed sign.

Save a study file and worksheet, then reopen them to confirm that the calculation and explanation can be reproduced.

Equilibrium voltage and operating voltage

The lab assumes reversible electrodes and zero liquid-junction potential.

It does not calculate current, kinetic overpotential, ohmic loss, mass-transfer limits or full electrolyte speciation.

A nonzero open-circuit potential is compatible with local electrode equilibrium; complete cell-reaction equilibrium gives $E=0$.

Exit question and course synthesis

A simulation converges and every balance closes. What could still make the predicted voltage or phase state wrong?

Explain the chain connecting chemical potential, model assumptions, equilibrium constraints, numerical checks and experimental evidence.

Independent practice • suggested 60–90 minutes

Assessment set 6 and rubric • Offline package

Two concentration-cell states, a reversal check and a reaction-scaling check, with consistent reference electrodes.

Retain the calculator export, your worksheet, a comparison plot/table and one independent check.
State an assumption that limits your conclusion.

Use the core labs on the [learning path](#). Optional extensions are additional work.

References and further study

Module reference deck · Lab sources and equations

IUPAC: standard electromotive force and standard equilibrium constant.

Synthetic worked examples illustrate calculations; they are not evidence of real-system accuracy.